



Capstone Kickoff

Everything you need to know: tracks, teams, iterations, deadlines, shadowing, submissions.

Project session 1 · 2026-09-29, 10:00–12:00 · T63 (Theorellsalen)

What the capstone is

- 40 of the course's 100 points — 30 team + 10 individual
- In teams of 2–3, build a full pipeline on real data: load → preprocess → features → leakage-safe validation → model → evaluation
- Runs agile, not waterfall — 4 iterations, each with its own milestone, so problems surface early instead of in week 9
- Ends with a showcase (2026-12-10) plus a team report and predictions.csv

This year's track pool

Track	Split unit	Honest baseline	Status
Sleep-EDF (staging)	subject	$\kappa = 0.835$ (LOSO)	Default
ECG CinC-2017 (rhythm)	record	macro-F1 = 0.655	Default
EMG Ninapro (gesture)	subject	0.78 within-subj / 0.19 new-subj	Default
BCI EEGMMIDB (motor imagery)	subject	≈ 0.51 , near chance without CSP*	Opt-in — sign-off required
HAR (IMU)	subject	too easy for this cohort (★★)	Not offered this year
CTG (fetal)	record	$\kappa \approx 0.11$ — a floor, not a challenge	Not offered this year

The 3 default tracks land closest together on the honest baseline metric, not just the nominal ★★★ label — that's why they're comparable.

*CSP = Common Spatial Patterns, a spatial filter that combines EEG channels to maximize the power difference between motor-imagery classes — the fix for BCI's near-chance baseline.

The 4 tracks, in detail

Track	Signal → task	Split unit / eval modes	Submission format
Sleep-EDF	Multi-ch PSG (EEG+EOG+EMG) → 5-class sleep stage, 30-s epoch	subject (LOSO)	record,epoch,label — 1 row / epoch
ECG CinC-2017	Single-lead ECG, 300 Hz → 4-class rhythm (Normal/AF/Other/Noisy)	record	record,label — 1 row / recording
EMG Ninapro	10-ch surface-EMG envelope → 12-class hand gesture	subject — BOTH within- & new-subject modes required	record,epoch,label — 1 row / 200-ms window
BCI EEGMMIDB (opt-in)	64-ch EEG → binary left/right motor imagery	subject — both modes required	record,epoch,label — 1 row / trial

Sleep-EDF: scaffolded, not easy

Easier to start

It's the book's own Ch16 worked example — the most scaffolding of any track, and BACKGROUND_MAP.md gives it the most course-section anchors of any track

The supplied baseline is the easiest of the three defaults to get running — you're largely adapting what Ch16 already walked through

Harder to do honestly

The baseline's real measured $\kappa = 0.835$ — above the human inter-rater ceiling ($\kappa \approx 0.76$). That's a warning sign, not a win: Wake dominates ~69% of epochs, so a mediocre pipeline still posts an inflated number

N1 stays a hard class (~0.20 recall) even with a working pipeline — and because the book already solved this exact problem, Criterion 7's bar is higher here: reproducing Ch16 is not improvement

For the 5 teams on Sleep-EDF (T1, T7, T9, T13, T14): don't trust the headline number — check the confusion matrix and per-class recall before you report a win.

Your track, your team, your shadow partner

- Your team ranked its top 2 of the 3 default tracks via the Canvas survey — the course team assigned one track each, balancing preference against roughly even counts per track
- At the same time, every team was paired with exactly one other team on a different track — your shadow partner for the rest of the course (never re-paired)
- Both are published on Canvas before today — check your track and your shadow partner now if you haven't
- Want BCI instead of your assigned default track? Talk to Farhad — opt-in, sign-off required

Teams, tracks & shadow pairs

Team	Track	Shadow partner	Partner's track
T1	Sleep-EDF	T3	ECG CinC-2017
T7	Sleep-EDF	T8	ECG CinC-2017
T9	Sleep-EDF	T5	EMG Ninapro
T10	ECG CinC-2017	T2	EMG Ninapro
T12	ECG CinC-2017	T4	EMG Ninapro
T13	Sleep-EDF	T6	EMG Ninapro
T14	Sleep-EDF	T11	EMG Ninapro

Each row is one shadow pair — always cross-track, so the comparison in your critical-comparison essay is between genuinely different signals. Every team appears exactly once. Track assignment: Sleep-EDF 5, ECG CinC-2017 4, EMG Ninapro 5.

Team roles (2–3 people — double up as needed)

Role	Owns	ClickUp tags
Project Manager	Workspace setup, task creation, integration, talking to the instructor	#management #integration
Preprocessing Lead	Signal cleaning, artifact handling, filter design, data validation	#preprocessing #filters
Feature Engineer	Feature extraction/selection, statistical checks, documentation	#features #analysis
ML Engineer	Model implementation, validation (correct split!), evaluation	#ml #classification

On a 3-person team the PM usually also carries one technical role. On a 2-person team, split all four in half and say so explicitly in ClickUp and your individual contribution disclosure.

Before you write a line of code

Set up your ClickUp workspace and assign the four roles — this becomes your task-assignment log

Book at least one recurring meeting per week

Agree how you'll communicate day to day, and where design decisions get written down

Plan the workload around your team's mix of backgrounds, not evenly by default — teams are built to mix on purpose

Why agile, not waterfall

Aspect	Waterfall	Agile (this project)
Feedback	Only at the end	After each iteration
Risk	High — all-or-nothing at the deadline	Low — incremental, caught early
Testing / validation	Discovered late, often too late to fix	The leakage-safe split is built and checked from iteration 1
Learning	Limited until the very end	Continuous — each iteration informs the next design choice

This mirrors the textbook's own §16.3 principle: baseline first, earn complexity later. It's also why rubric Criterion 9 grades your iteration & revision history directly.

The 4 iterations, at a glance

#	Focus	Milestone
1 — Baseline pipeline	default_baseline() runs end to end, smoke-tested	2026-10-06
2 — Real DSP & features	Preprocessing + feature construction replace the placeholders	2026-10-15 → 10-26
3 — Honest validation	Correct split unit enforced in code; both eval modes where required	2026-11-10
4 — Polish & delivery	Reproducibility, report, predictions.csv, showcase prep	2026-11-24 → 12-01

Track-agnostic build order — applies to Sleep-EDF, ECG, EMG, or the BCI opt-in alike. These are milestones, not Canvas submissions — tracked in ClickUp, checked in project sessions.

Today: Iteration 1 — Baseline pipeline

- Milestone extended one week: now 2026-10-06, not today — use the rest of this session as a head start, not a deadline to hit before you leave
- “Done” means: `adapter.default_baseline()` runs end to end on your track, and the track's `smoke()` path passes
- This is the supplied baseline — you are not writing your own pipeline yet, just getting the shipped one running on your track's real data
- Read your track's card + instructions in `tracks/` first — dataset, license, split unit, and the literature-review anchors are all there

Iterations 2–4 — what “done” looks like

Iteration	What “done” looks like
2 — Real DSP & features (due 2026-10-26)	Your own preprocessing and feature construction have replaced the baseline's placeholders — this is where the DSP marks are
3 — Honest validation (due 2026-11-10)	The correct split unit is enforced in code everywhere; EMG/BCI report both within-subject and new-subject modes side by side
4 — Polish & delivery (due 2026-12-01, before the 2026-12-10 showcase)	Fixed seed, pinned environment, smoke() still passes; report drafted; predictions.csv ready; showcase slides prepped

Every date, in one place

Date	What it is
2026-10-06	Iteration 1 — Baseline milestone
2026-10-15 → 2026-10-26	Iteration 2 — Real DSP & features milestone
2026-11-10	Iteration 3 — Honest validation milestone
2026-11-24 → 2026-12-01	Iteration 4 — Polish & delivery milestone
2026-12-10, 08:00–12:00	Showcase (T53, Hertzsalen)
2026-12-10	Report + predictions.csv due — 30 pts (team)
2026-12-17	Critical-comparison essay due — 5 pts (individual)
2026-12-17	Teamwork reflection due — part of 5 pts (individual)
2026-12-17	Peer evaluation due (private) — part of 5 pts (individual)

Iterations 1–4 are milestones — tracked in ClickUp, not separately submitted on Canvas. Everything below the showcase is a real graded Canvas deadline. If any other page disagrees with these dates, this list is what counts.

ClickUp setup (Project Manager's job, today)

Workspace name: CM2013_<Track>_Group[X] — e.g. CM2013_SleepEDF_Group3. Add the instructor as viewer.

One sprint folder per iteration, named with its real milestone date.

Tags: priority (●/●/●), your track's signal (#EEG/#ECG/#IMU/#EMG), status (#BLOCKED #NEEDS-REVIEW #BUG), component (#preprocessing #features #classification #documentation).

Task titles: [Component] Specific action — e.g. “[Preprocessing] Notch filter for 50 Hz mains”. Status flow: To Do → In Progress → Review → Testing → Complete.

Sprint discipline

Every iteration

- Sprint planning day 1: PM creates tasks, team assigns and tags them
- Daily standups as ClickUp comments: "Today / Tomorrow / Blockers"
- Mid-sprint: PM scans for #BLOCKED, redistributes work
- Weekly: PM updates progress, checks time estimates vs. actuals

Red flags the PM must monitor

- ⚠ A task untouched for 48h
- ⚠ Multiple #BLOCKED tags
- ⚠ Uneven task distribution across members
- ⚠ No completed tasks by day 3 of an iteration
- ⚠ Missing standups

These red flags are exactly what individual teamwork grading is read from — keeping ClickUp honest makes your own grading easier and fairer.



Shadow pairing

- Every team is paired, once, with exactly one other team on a different track — never re-paired, and always cross-track on purpose (so you compare across genuinely different signals/tasks, not two teams doing the same thing)
- Published on Canvas at the same time as track assignments — check who your shadow partner is now, alongside your track
- Look in on their project informally around the project's mid-point (~Iteration 2/3) — this is a check-in, nothing to submit yet
- The graded output from this pairing is the critical-comparison essay, written after the showcase — see the next slide for exactly what and how to submit

3 individual submissions

What	Format	Counts as
Critical-comparison essay (about your shadow partner)	300–500 words. What's different in their approach, what would you borrow, what would you change about your own team's, having seen theirs	5 pts, direct
Teamwork reflection	150–300 words. One disagreement or workload imbalance and how it was handled — or say your team had none, and how you kept it that way	Part of the 5-pt teamwork score
Peer evaluation (private)	Rate each teammate + yourself, 1–5, on 3 criteria, one justifying sentence each — only the instructor/TA ever sees it	Part of the 5-pt teamwork score

All three are submitted individually on Canvas, all due 2026-12-17. The teamwork score is never graded from the reflection alone — it's reflection + peer evaluation + your ClickUp task log, combined.

Grading: 30 team + 10 individual points

#	Criterion	Pts
1	Pipeline integrity, 7 separable stages, load through report	3
2	Signal processing rigor, real DSP, justified in the report	5
3	Validation honesty, correct split unit, full metric panel	6
4	Reproducibility & engineering, seeded, pinned, reruns clean	2
5	Defence of the report's decisions (oral)	3
6	Deadlines	1
7	Improvement on the baseline, evidence-based comparison	4
8	Track-specific execution check (varies by track)	3
9	Iteration & revision history, the log, not just the result	3

Same 9 criteria and point values across all 4 tracks — only Criteria 7–8 are concretized per track. Individual 10 pts (essay + teamwork reflection) is separate.

Two things to unlearn right now

✗ "We must beat the supplied baseline."

✓ Criterion 7 grades the quality of your evidence — controlled comparisons, results read with their spread, alternatives genuinely tried. A well-justified decision to keep a lower-scoring pipeline earns full marks.

⚠ "A group-aware split is a nice-to-have."

🚫 Criterion 3 is a hard fail: no group-aware split in the code is a hard fail on that criterion regardless of the reported number — not a partial-credit issue.

Right now: for the rest of this session

1. Confirm your assigned track and your shadow partner on Canvas
2. Set up your ClickUp workspace and assign the four roles
3. Read your track's card + instructions in tracks/ on GitHub
4. Get `adapter.default_baseline()` running on your track — aim to leave today with it smoke-tested, or well underway before the 2026-10-06 milestone
5. Book your team's weekly recurring meeting

Key resources

- github.com/farhad-abtahi/CM2013 — tracks/ folder: cards, instructions, adapter, the working baseline for every track
- Canvas: Pick your track · Ground rules · Project Management & ClickUp · Capstone Assessment Rubric · m4-module-overview (the definitive date list)
- `tracks/START_HERE.md`, `results_log_TEMPLATE.md`, `HOLDOUT_EVALUATION.md`
- Stuck? Raise it inside your team first, directly — then bring it to a project session or message the instructor



Let's get started

Track picked. Team formed. ClickUp open. Baseline next.